Experiment No. 6

Aim: Apply K-Means and Hierarchical Clustering on sample datasets

Theory

1. Unsupervised Learning & Clustering

Clustering is an unsupervised machine learning technique that groups data points into subsets (clusters) such that points within the same cluster are more similar to each other than to points in other clusters. Unlike supervised learning, clustering algorithms work without labeled training data, making them ideal for exploratory data analysis and pattern discovery.

Clustering is widely used in customer segmentation, anomaly detection, document grouping, image compression, and bioinformatics.

2.2 K-Means Clustering

K-Means is one of the most popular and computationally efficient clustering algorithms. It partitions n observations into k clusters, where each observation belongs to the cluster with the nearest centroid (mean).

Key characteristics:

- Requires the number of clusters k to be specified in advance
- Assumes clusters are convex and isotropic (spherical in shape)
- Sensitive to initial centroid placement (mitigated by K-Means++ initialization)
- Guaranteed to converge but may reach a local minimum
- Time complexity: O(n * k * t * d) where t = iterations, d = dimensions

2.3 Hierarchical Clustering

Hierarchical clustering builds a tree-like structure (dendrogram) of nested clusters. It does not require specifying k in advance — the number of clusters is determined by cutting the dendrogram at a chosen height.

Two main strategies:

- Agglomerative (bottom-up): Each point starts as its own cluster; pairs are merged iteratively — used in this experiment
- Divisive (top-down): Starts with one cluster and recursively splits

Linkage criteria determine how the distance between clusters is calculated. Ward linkage (used here) minimizes the total within-cluster variance at each merge step, making it robust for compact, well-separated clusters.

3. Steps / Methodology

Step 1: Import Libraries

The following Python libraries are used:

- numpy, pandas — data manipulation
- scikit-learn — KMeans, AgglomerativeClustering, StandardScaler, metrics
- scipy — linkage matrix and dendrogram for hierarchical clustering
- matplotlib — all visualizations

Step 2: Load & Explore Dataset

The Mall Customer Segmentation dataset contains 150 records with the following features:

Feature	Type	Description
annual_income_k	Numeric	Annual income in thousands ($)
spending_score	Numeric	Mall spending score (1–100)
age	Numeric	Customer age in years
purchase_frequency	Numeric	Number of purchases per month

Step 3: Preprocessing

All features are standardized using StandardScaler before clustering to ensure equal contribution from each feature regardless of its original scale.

X_scaled = (X - mean) / std_deviation

This transforms each feature to have mean = 0 and standard deviation = 1.

Step 4: Finding Optimal k

Three methods are used to determine the best number of clusters:

- **Elbow Method:** Plot inertia (within-cluster sum of squares) vs k; look for the 'elbow' where improvement flattens
- **Silhouette Analysis:** Compute silhouette score for k = 2 to 10; choose k that maximizes the score
- **Davies-Bouldin Score:** Lower is better; used as a cross-validation metric

Result: Optimal k = 7 was selected based on the maximum silhouette score.

Step 5: Apply K-Means Clustering

K-Means is applied with the following configuration:

- n_clusters = 7 (optimal k)
- n_init = 10 (runs 10 times with different centroids, picks best)
- max_iter = 300 (maximum convergence iterations)
- random_state = 42 (reproducibility)

After fitting, cluster labels are assigned to each data point, and centroids are inverse-transformed back to the original scale for interpretability.

Step 6: Apply Hierarchical Clustering

Agglomerative Hierarchical Clustering is applied with Ward linkage:

- A linkage matrix Z is computed using scipy's linkage() function
- AgglomerativeClustering with n_clusters=7 and linkage='ward' assigns final labels
- The dendrogram is visualized with a horizontal cut line at the height corresponding to k=7

Step 7: Evaluate Both Models

Three evaluation metrics are computed for both algorithms and compared side by side.

Step 8: Visualize Results

An 8-panel visualization is produced including:

- Elbow curve

- Silhouette bar chart
- Davies-Bouldin line plot
- K-Means scatter plots (Income vs Spending, Age vs Income)
- Hierarchical scatter plot
- Dendrogram
- Metric comparison bar chart

Algorithm Recommendation

Use Case	Recommended Algorithm	Reason
Large datasets	K-Means	Faster, O(nkt) complexity
Exploratory analysis	Hierarchical	Dendrogram shows all k levels at once
Unknown k	Hierarchical	No need to pre-specify k
Marketing segmentation	K-Means	Better silhouette, cleaner segments

Code:

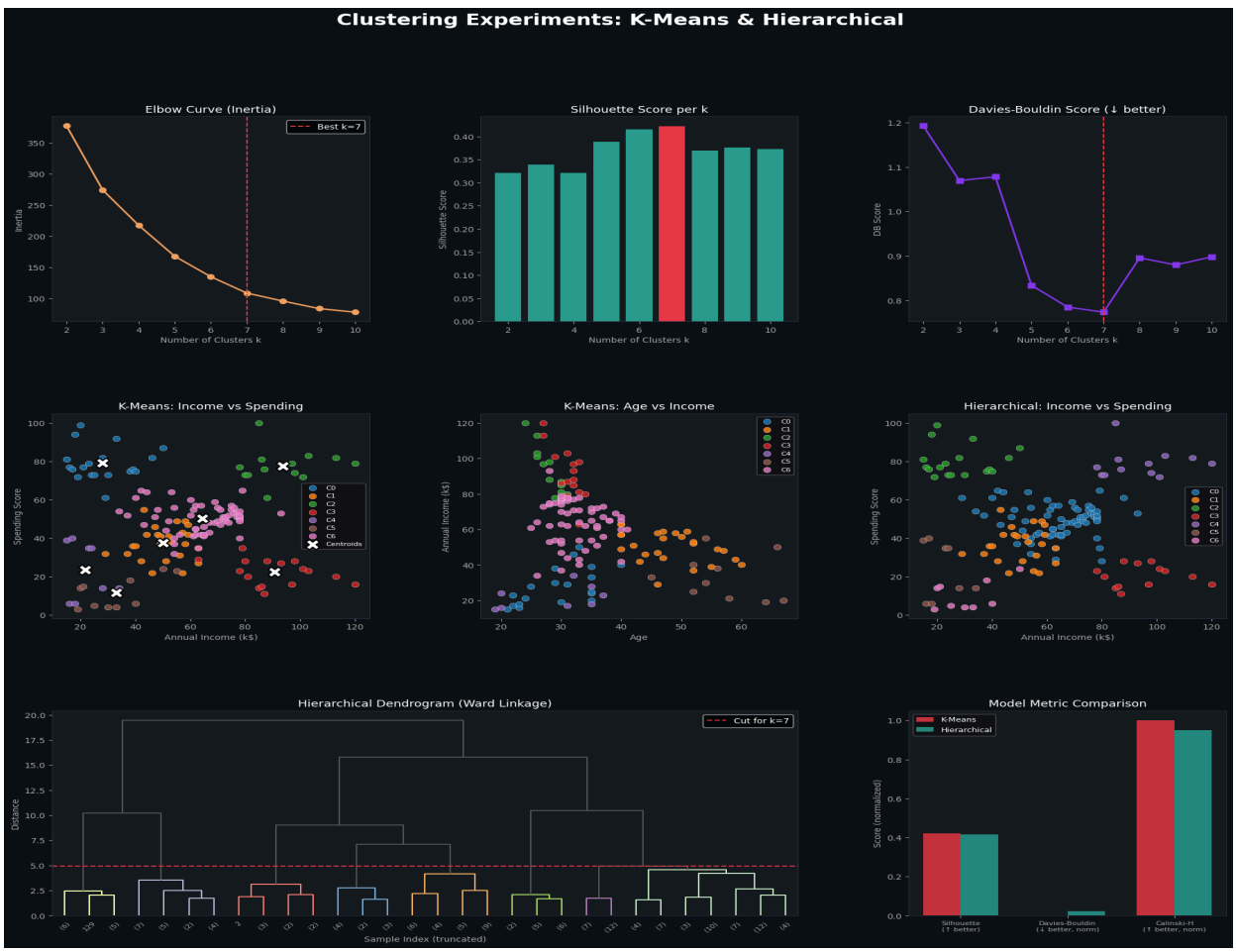
MLDL Lab
All code present in this Colab notebook

Output:

```
 SUMMARY
================================================================
  Dataset          : 150 Mall Customers, 4 features
  Optimal k        : 7

  K-Means
    Silhouette     : 0.4218
    Davies-Bouldin: 0.7734
    Calinski-H     : 107.40

  Hierarchical (Ward)
    Silhouette     : 0.4162
    Davies-Bouldin: 0.7568
    Calinski-H     : 101.96
```



Conclusion: I have successfully implemented and understood K mean and Hierarchical clustering